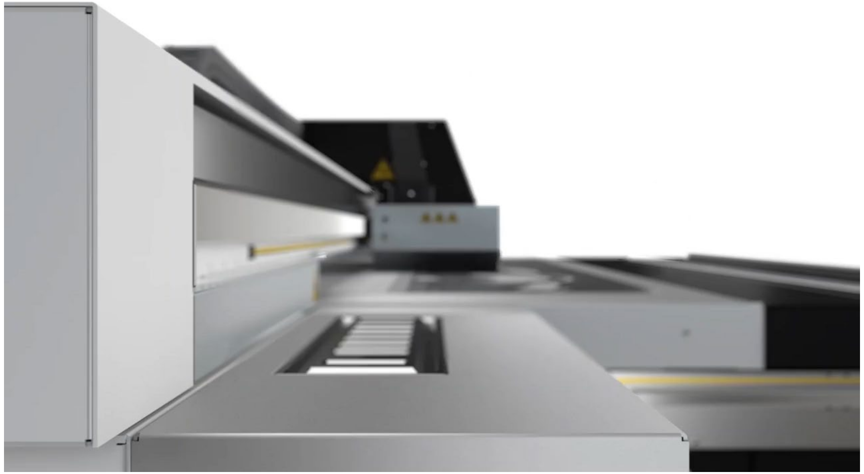
Learning Control Challenge

2025-2026

Mechanical Engineering, Control Systems Technology

EINDHOVEN UNIVERSITY OF TECHNOLOGY

Introduction



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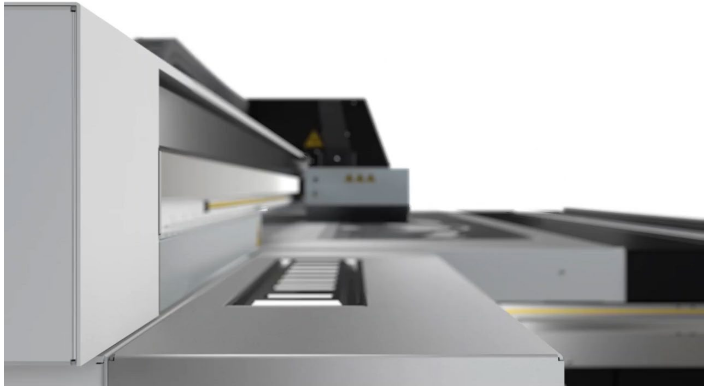

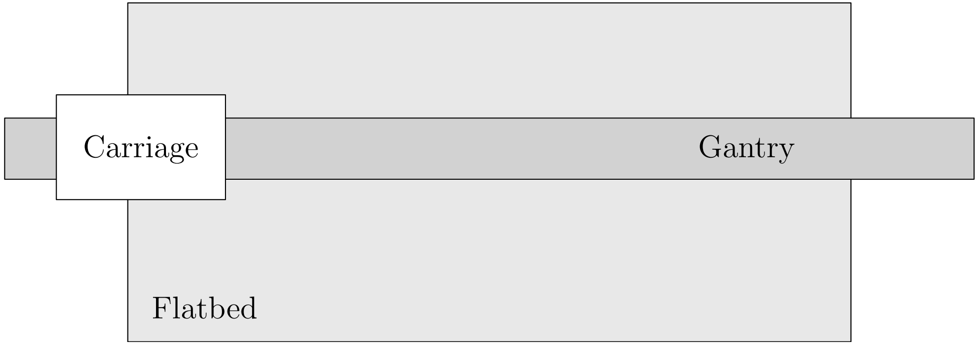

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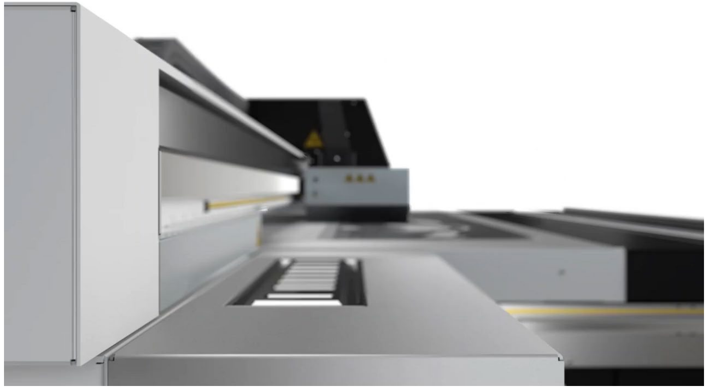

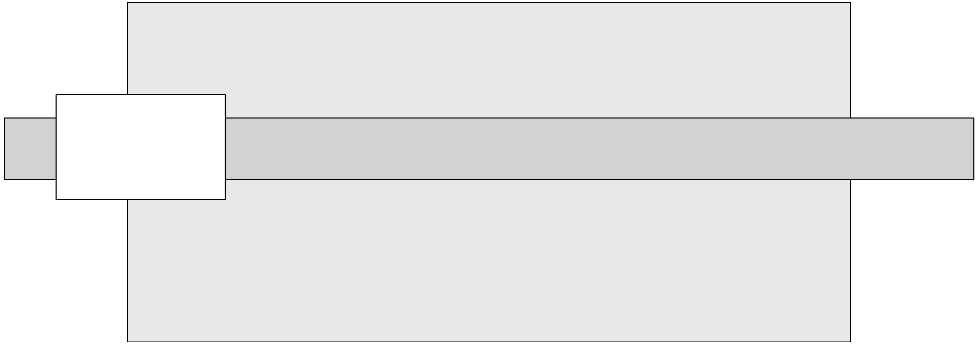

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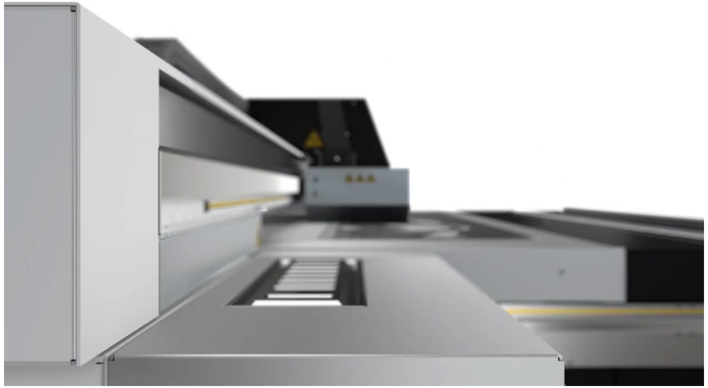

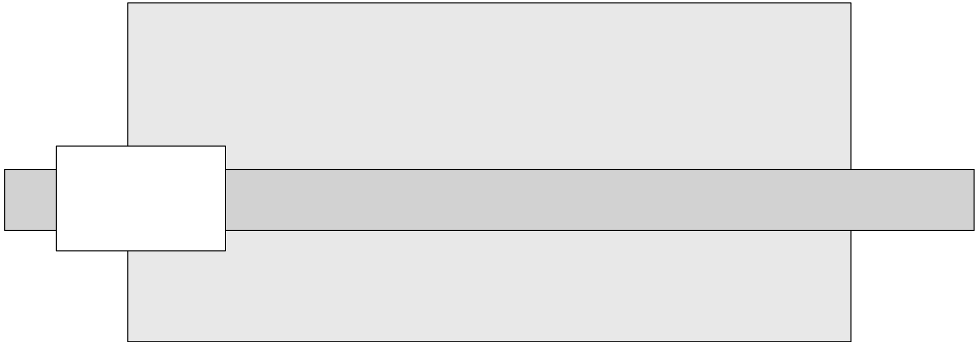

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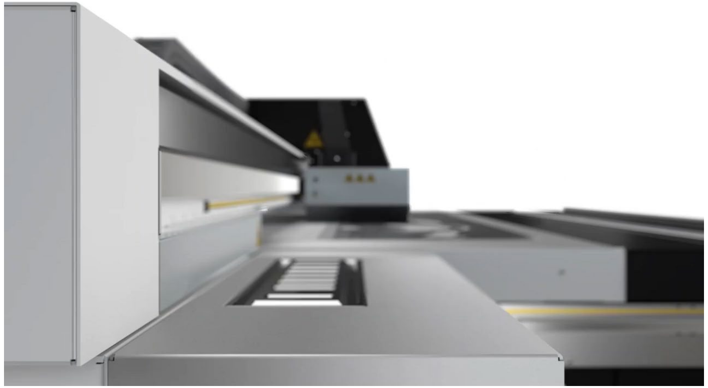

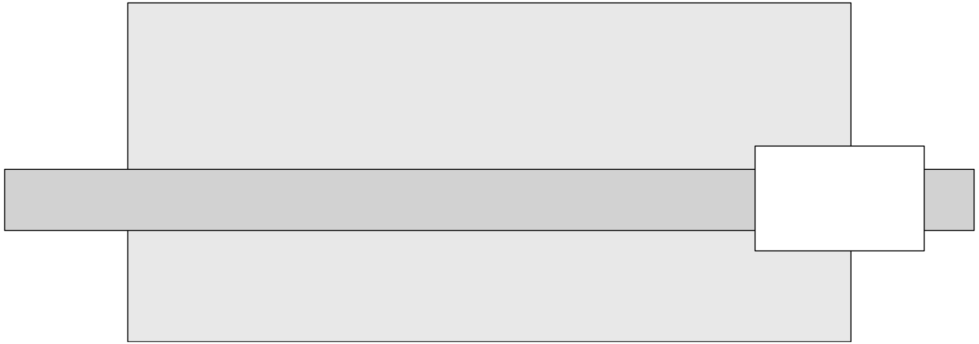

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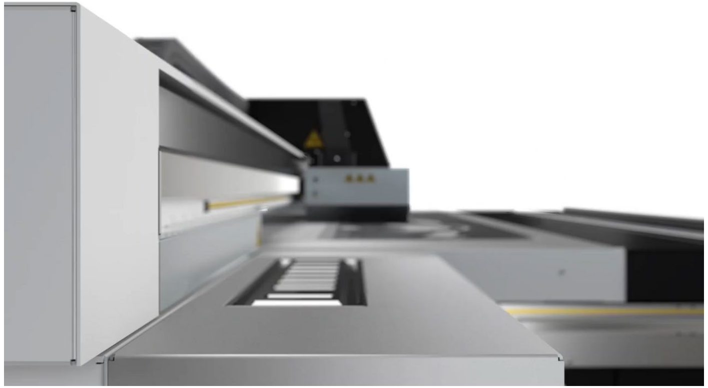

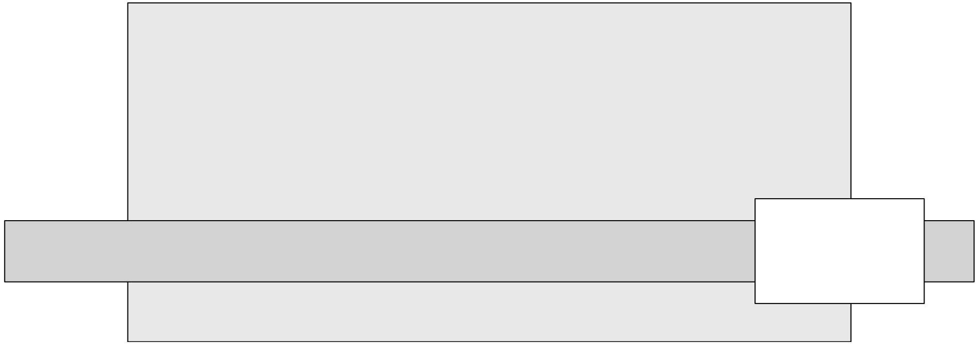

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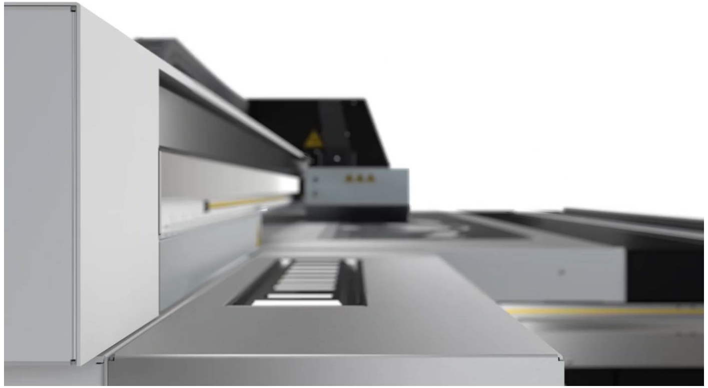
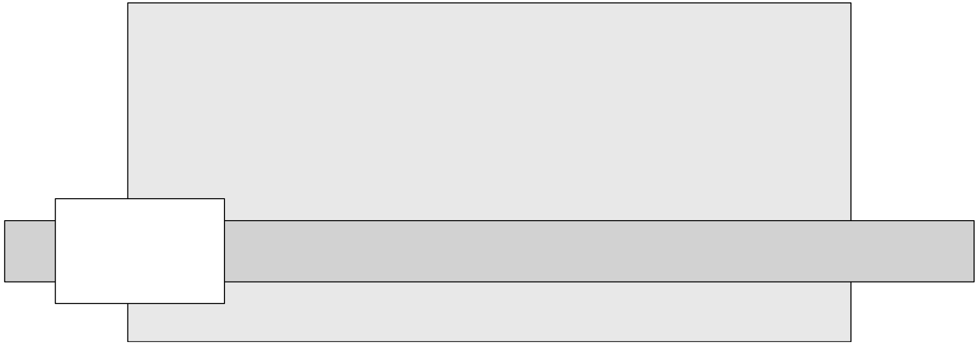
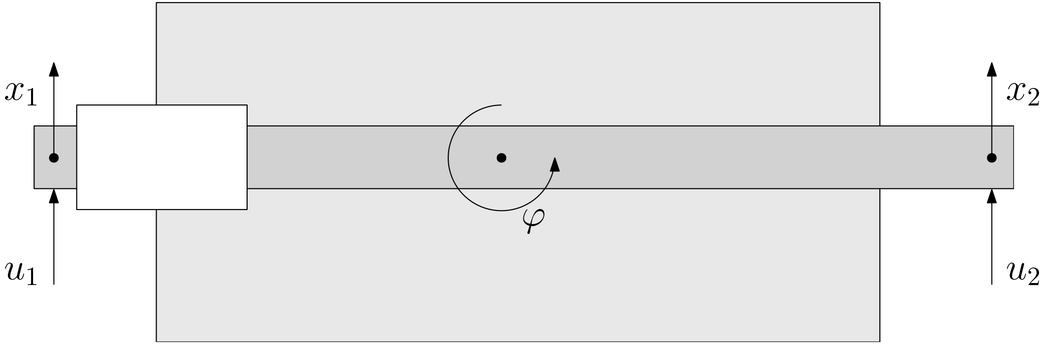


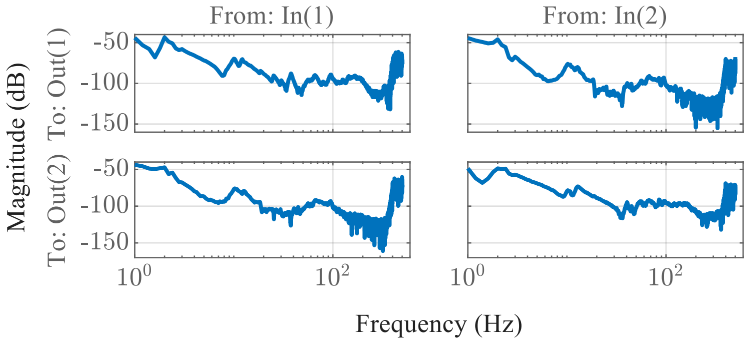


Introduction

- Two-input two-output system

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = P \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

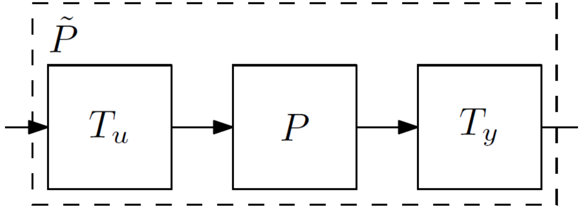




Introduction

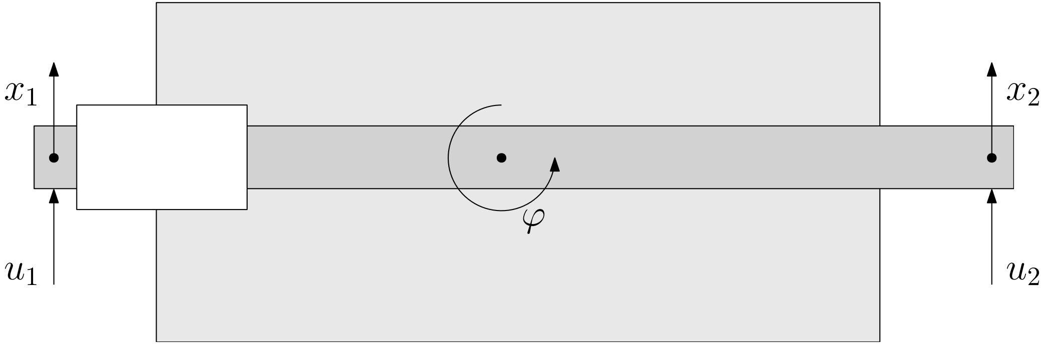
- Two-input two-output system

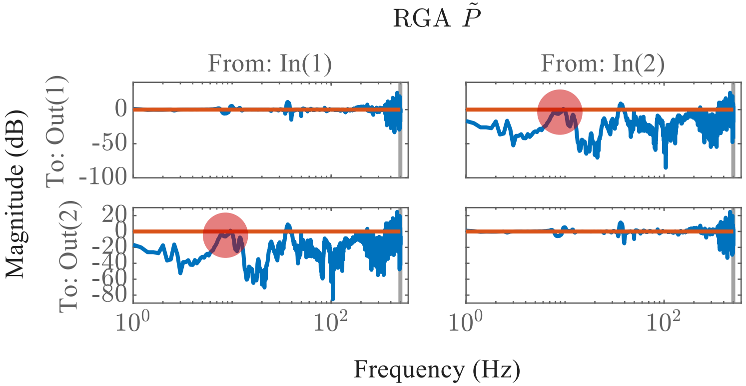
$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = P \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

- Partially decoupled
  - 

$$\begin{bmatrix} x \\ \varphi \end{bmatrix} = \tilde{P} \begin{bmatrix} u_x \\ u_\varphi \end{bmatrix}$$

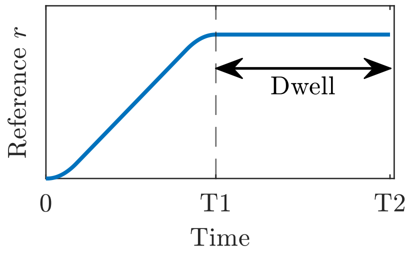
  - RGA shows interaction
- What happens when applying a step in x?



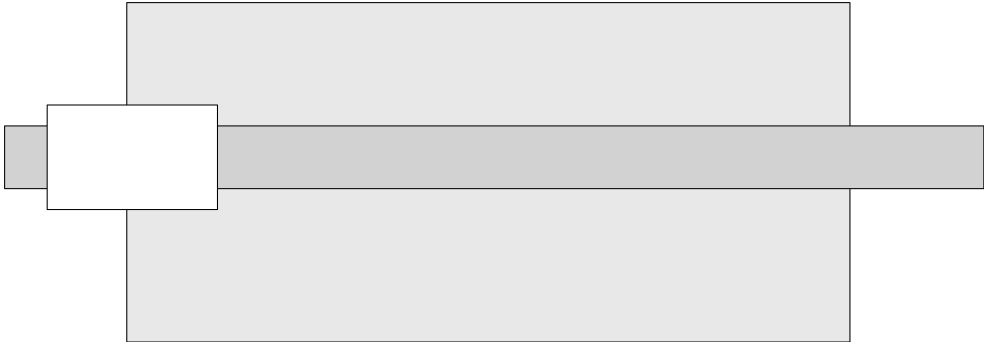


Introduction

- Fast reference in x-direction:

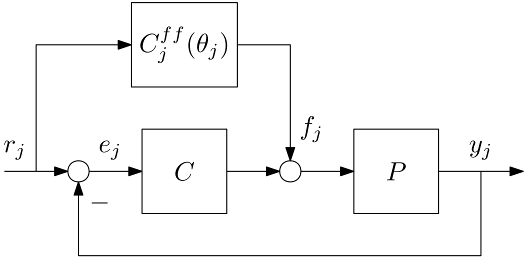


  - Reference might change



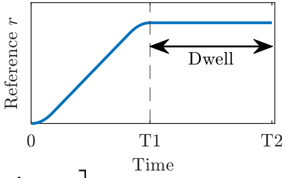
- **Result:** settling errors in both x and φ directions
  - Carriage has to wait… printing speed limited
- **Problem:** how to <u>minimize</u> settling errors in <u>both directions</u> x and φ for a <u>rational</u> MIMO motion <u>system</u>?

Recap: ILC with basis functions

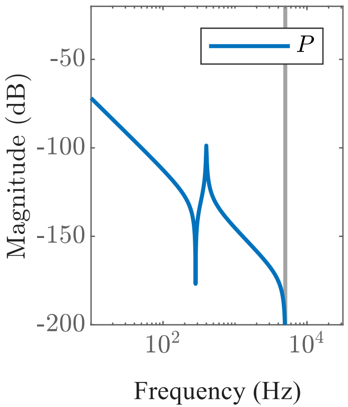


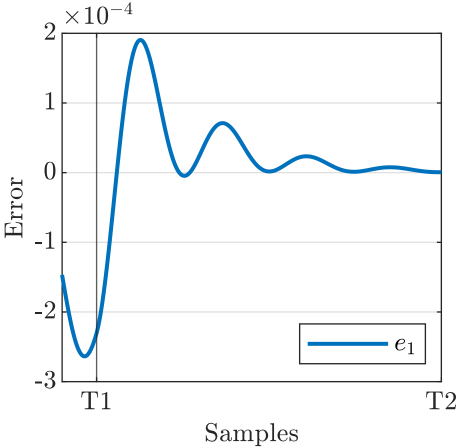
- Control setup:
  - ▶ Tracking error $e_j = S(1 - PC^{ff})r_j$
  - ▶ Design $C^{ff} = P^{-1}$ to obtain $e_j = 0$ (for any task)
- Optimal ILC synthesis
  - ▶ Performance criterion: $\mathcal{J}(\theta_{j+1}) = \|\hat{e}_{j+1}(\theta_{j+1})\|^2_{W_e} + \|\Psi(r_j)\theta_{j+1}\|^2_{W_f} + \|\Psi(r_j)(\theta_{j+1} - \theta_j)\|^2_{W_{\Delta f}}$
  - ▶ Substituted: $f_j = C_j^{ff}(\theta_j)r_j = \Psi(r_j)\theta_j$

- Example (SISO):
  - ▶ 4th order motion system (collocated)
  - ▶ Apply reference:

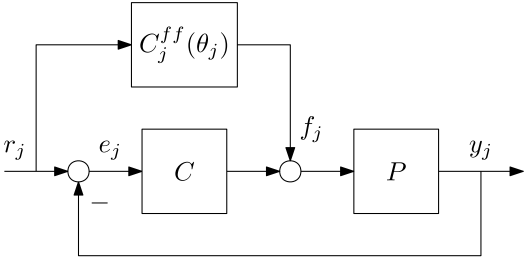


  - ▶ Pick $\Psi(r) = \begin{bmatrix} a & j & s \end{bmatrix}$





Recap: ILC with basis functions



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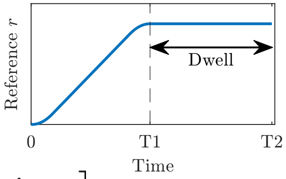
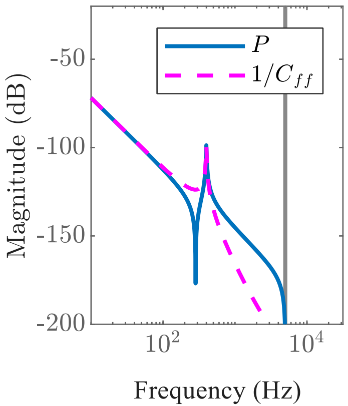
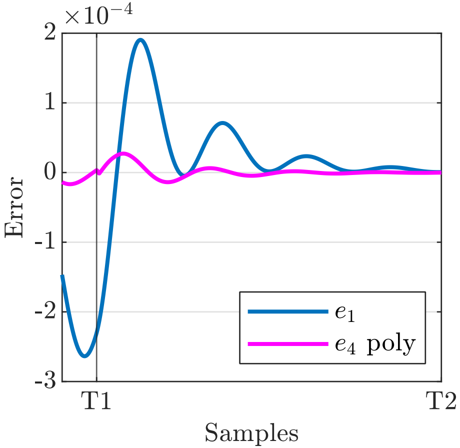


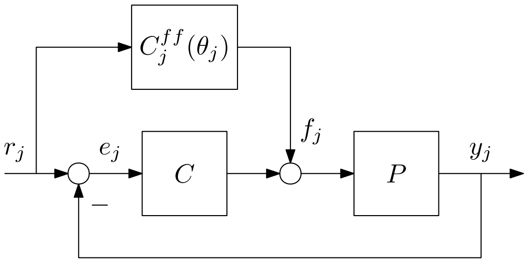
  - ▶ Pick $\Psi(r) = \begin{bmatrix} a & j & s \end{bmatrix}$
  - ▶ Polynomial ffw only able to capture poles P !
  - ▶ Lets fix that!





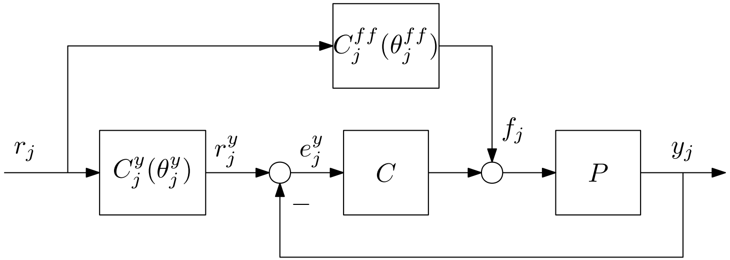
ILC with basis functions and input shaping

- Control setup:



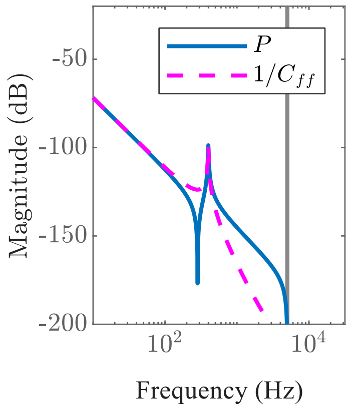
ILC with basis functions and input shaping

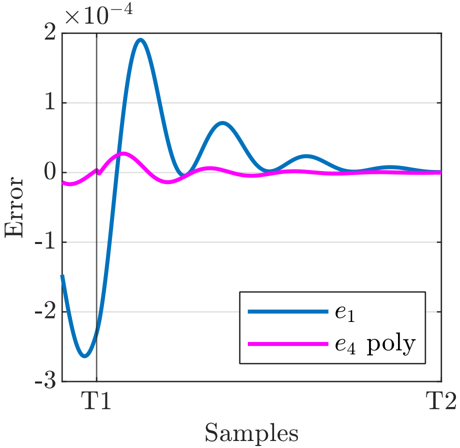
- Control setup:
  - Tracking error $e_j^y = S\left(C_j^y - PC_j^{ff}\right) r$
  - Design $\frac{C^{ff}}{C^y} = P^{-1}$ to obtain $e_j^y = 0$ (for any task)



- Optimal ILC synthesis:
  - Performance criterion: $\mathcal{J}(\theta_\Delta) = \underbrace{\|\hat{e}_{j+1}^y(\theta_\Delta)\|_{W_e}^2}_{\text{error}} + \underbrace{\|\Psi^{ff}(\theta_\Delta^{ff} + \theta_j^{ff})\|_{W_f}^2 + \|\Psi^{ff}(\theta_\Delta^{ff})\|_{W_{\Delta f}}^2}_{\text{feedforward}} + \underbrace{\|\Psi^y(\theta_\Delta^y + \theta_j^y)\|_{W_{r_y}}^2 + \|\Psi^y(\theta_\Delta^y)\|_{W_{\Delta r_y}}^2}_{\text{input shaper}}$
  - Problem linear in θ_Δ ☺

- Example (cont.)
  - Pick $\Psi^{ff}(r) = \begin{bmatrix} a & j & s \end{bmatrix}$
  - Pick $\Psi^y(r) = \begin{bmatrix} v & a & j & s \end{bmatrix}$





TU/e

ILC with basis functions and input shaping

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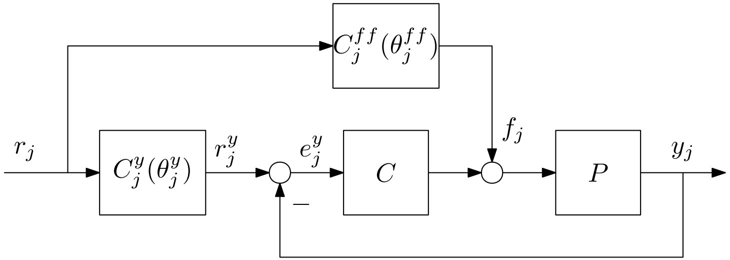

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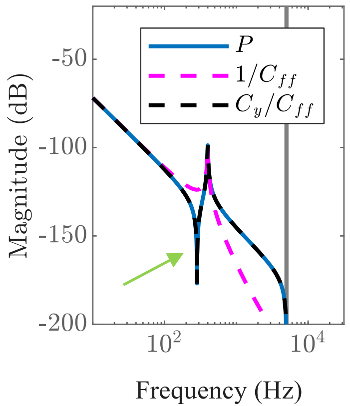
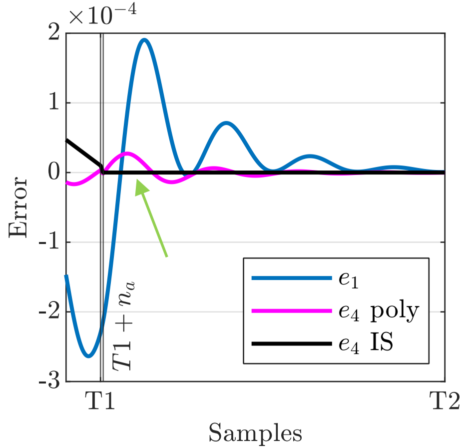
- Example (cont.)
  - Pick $\Psi^{ff}(r) = \begin{bmatrix} a & j & s \end{bmatrix}$
  - Pick $\Psi^y(r) = \begin{bmatrix} v & a & j & s \end{bmatrix}$
  - High performance! Poles and zeros covered
  - Let order C^y be n_a
    - Error $e(k) = e^y(k)$ for $k \in [T1 + n_a,\ T2]$





TU/e

The Challenge

- **Goal:** minimize settling errors in both x and φ directions by iteratively learning a MIMO feedforward controller and input shaper

- Feedforward C^{ff} and input shaper C^{y} should fulfill:
  - R1: settling errors e^{x} and e^{φ} minimized
  - R2: controllers robust to reference r^{x} changes
  - R3: controllers must be able to deal with nonlinear effects, e.g., friction

TU/e

The Challenge

- **Goal:** minimize settling errors in both x and φ directions by iteratively learning a MIMO feedforward controller and input shaper

- Possible research directions:
  1. **MIMO ILC with BFs and input shaper:** investigate whether MIMO approach necessary to solve the problem, if yes, develop MIMO synthesis framework.
  2. **Parameterization:** what structure and basis functions do you need in the controllers?
  3. **Non-linear compensation:** how to deal with non-linear friction effects?
  4. **Robustness:** can you ensure robustness of your ILC framework?
  5. **Convergence:** can you develop theoretical guarantees for convergence of your algorithm?

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The Challenge

	C^{ff}	C^{ff} and $C_j^y = z^{-d}$	C^{ff} and $C_j^y = 1 + \sum_{i=1}^{n_a} \left(\frac{1-z^{-1}}{T_s}\right)^i \theta^i$
SISO			
MIMO			

The Challenge

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  5. **Convergence:** can you develop theoretical guarantees for convergence of your algorithm?
  6. **Be creative** ☺ feel free to go beyond ideas presented today.
- Remember, we don't know the answers either and are willing to think along with you!

Challenge Deliverables

- Deliverables:
  - 10-minute midterm presentation (informal). Friday May 29 (see schedule).
  - MIMO feedforward and input shaper design procedure and Matlab/Simulink files to generate the controllers.
  - Report on the approach in paper style, template will be provided. Use scientific approach!
  - A short movie explaining your approach (±5 min.). Use your creativity ☺
- Criteria:
  - Performance
  - Creativity
  - Scientific basis
  - Report
  - Movie

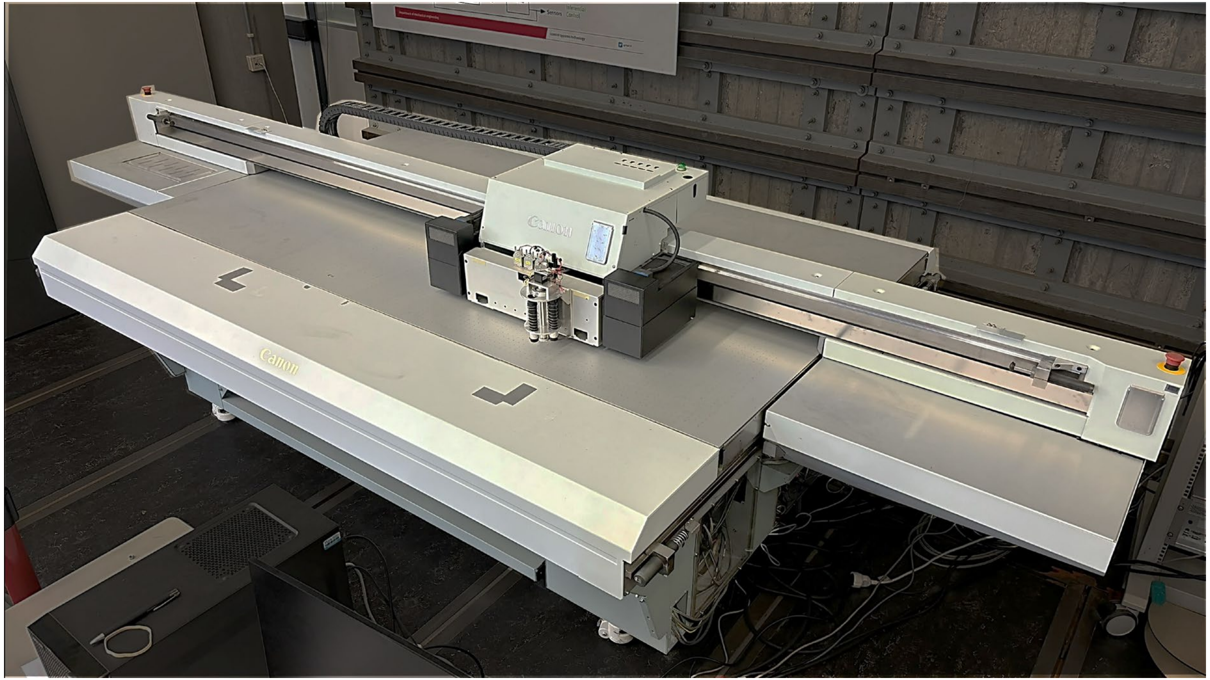
Next steps for the course

- Week-to-week planning: see Challenge_Description.pdf (Canvas)

From now on:

- One meeting with your TA per week, preferably in LC timeslots. (Discuss with your TA).
- You'll receive an example Matlab script that simulates a SISO system for ILC with basis functions and input shaper.
- You'll receive non-parametric models of the Arizona. Parametric models will follow later.
- TA is present when you do Arizona experiments.
- In week 6, we'll have a session with short presentations on the initial ideas.
- We'll mainly communicate via Canvas and via TA's. Please discuss problems/unclarities with your TA.
- Final examination date: Wednesday July 1st (9:00-11:15)

Meet with your TA, and... Enjoy!



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Group 2: Gijs van Meerbeeck
Group 3: Timo de Groot
Group 4: Lennart Blanken*
Group 5: Tjeerd Ickenroth
Group 6: Maarten van der Hulst